

YUEZHEN (LILY) DONG

yuezhen-dong.com ◇ Ningmengchem@gmail.com ◇ github.com/Lilllilly06 ◇ linkedin.com/in/yuezhen-dong-b68281268/

EDUCATION

University of Waterloo – Faculty of Engineering

Expected Graduation May 2028

BASc in Computer Engineering

Waterloo, ON

Relevant Coursework: Data Structures & Algorithms, Digital Systems, Embedded Systems

WORK EXPERIENCE

Meta Engineering Fellow

Jun 2026 – Sep 2026

Python, Linux, Flask, Docker, Nginx, GitHub Actions, Prometheus/Grafana

Meta x MLH Fellowship

- Automated production deploys from **8 manual steps / 10–15 min to one merge / 3–5 min** by building test-gated GitHub Actions CI/CD for a **3-service Docker Compose stack** on DigitalOcean.
- Built **12-panel Prometheus/Grafana monitoring** with ~2-minute failure alerts; traced **stress-ng** jobs and a **24.6 GB log**, reducing root-disk utilization from **99% to 7%**.

Software Engineering Intern – Return Internship

May 2026 – Aug 2026

Python, PyTorch, TypeScript, React Native, Ruby on Rails, GRPO, NeMo-RL

Shopify

- Raised Sidekick Analytics subagent unit-test pass rate from **43.4% to 87%** by building an end-to-end agentic RL pipeline with GRPO/NeMo-RL, verifiable rewards, grouped rollouts, and held-out tool/judge evaluation.
- Built embedding-reranked RAG and prompt-orchestration infrastructure for agentic AI tooling, reducing irrelevant retrieved context by ~**40%** across developer workflows.
- Built Quick Device Activation for Shopify POS with BLE pairing, QR/manual-code fallback, and OAuth Device Authorization, cutting merchant device setup from ~**5 minutes to under 30 seconds**.

Software Engineering Intern

Sep 2025 – Dec 2025

React, TypeScript, Ruby on Rails, GraphQL

Shopify

- Owned a React/GraphQL staff-assignment workflow end-to-end, adding server-side pagination and query optimizations that reduced staff-search response time by ~**60%**.
- Migrated a critical Retail GraphQL workflow across **6 Shopify systems**, preserving backward-compatible data contracts and shipping with **zero production regressions**.

Web Developer Co-op

Jan 2025 – Apr 2025

Java, Spring Boot, Maven, Apache POI, Git

AGF Investments

- Built a Java/Spring Boot ScoreCard pipeline with Apache POI streaming and configurable transformations, automating XLSX reporting across **4 departments** and saving **20+ staff hours/week**.

PERSONAL PROJECTS

AsterFind  GitHub

Python, NLP, BM25, TF-IDF, Collaborative Filtering, Kroger API, Docker, GitHub Actions

- Built an LLM-assisted shopping search engine with JSON-Schema intent extraction, live Kroger catalogue retrieval, hard-constraint filtering, and hybrid BM25/TF-IDF plus personalized item-CF reranking.
- Reached **0.982 nDCG@5 / 0.976 MRR@5** on a reproducible retrieval benchmark; shipped provider fallbacks, Docker/CI, and **26 automated unit/E2E tests** to keep ranking behavior auditable.

Temporal Maze / Puppy Portal  GitHub

C++, Unreal Engine 5.8, Gameplay Framework, Actor Components, UInterfaces, Automation Testing

- Rebuilt my original Python/Pygame time-clone prototype as a native **UE5.8 C++** campaign across **3 trials**, with reusable components and fixed-rate shared-clock playback for synchronized echoes.
- Implemented portals, checkpoints, persistent SaveGame state, and signal-driven gates; verified deterministic gameplay with Unreal automation, headless smoke, and end-to-end integration tests.

RELEVANT SKILLS

Languages Python, C++, Java, JavaScript/TypeScript, C, Go, SQL, Ruby, Bash

ML / AI PyTorch, scikit-learn, Information Retrieval, NLP, RAG, Embeddings, LLM Agents, GRPO, NeMo-RL

Technologies React, React Native, FastAPI, Flask, Ruby on Rails, Spring Boot, GraphQL, Linux, Docker, CI/CD, Nginx, PostgreSQL/MySQL, Prometheus/Grafana